

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases}$$

$$|f(x, y_1) - f(x, y_2)| \leq k|y_1 - y_2|$$

$$\alpha \in [x_0 - \alpha, x_0 + \alpha]$$

$$\begin{cases} y' = f(x, y) + \delta(x) \\ y(x_0) = y_0 + \varepsilon \end{cases}$$

$$\text{not } y(x, \delta, \varepsilon) - \text{sol ec.}$$

$$\text{not } \|y(x, \delta, \varepsilon) - y(x)\|$$

$$y(x, \delta, \varepsilon) = \int_{x_0}^x f(t, y(t, \delta, \varepsilon)) dt + \int_{x_0}^x \delta(t) dt + y_0 + \varepsilon$$

$$y(x) = \int_{x_0}^x f(t, y(t)) dt + y_0$$

(+)

$$\Rightarrow \|y(x, \delta, \varepsilon) - y(x)\| = \left\| \int_{x_0}^x f(t, y(t, \delta, \varepsilon)) - f(t, y(t)) dt + \int_{x_0}^x \delta(t) dt + \varepsilon \right\| \leq$$

$$\leq \int_{x_0}^x |f(t, y(t, \delta, \varepsilon)) - f(t, y(t))| dt + \left| \int_{x_0}^x \delta(t) dt \right| + \varepsilon \quad \text{f. Lipschitziană}$$

$$\leq \int_{x_0}^x k |y(t, \delta, \varepsilon) - y(t)| dt + \int_{x_0}^x |\delta(t)| dt + \varepsilon$$

$$\|y(x, \delta, \varepsilon) - y(x)\| \leq k \|y(x, \delta, \varepsilon) - y(x)\| \alpha + \alpha \|\delta\| + \varepsilon$$

$$\|y(x, \delta, \varepsilon) - y(x)\| (1 - k\alpha) \leq \alpha \|\delta\| + \varepsilon \Rightarrow$$

$$\Rightarrow \|y(x, \delta, \varepsilon) - y(x)\| \leq \frac{1}{1 - k\alpha} (\alpha \|\delta\| + \varepsilon) \quad \text{pp } (1 - k\alpha) > 0 \Rightarrow \alpha < \frac{1}{k}$$

$$\Rightarrow \lim_{\substack{\delta \rightarrow 0 \\ \varepsilon \rightarrow 0}} y(x, \delta, \varepsilon) = y(x)$$

$$\delta \rightarrow 0 \\ \varepsilon \rightarrow 0$$

solutia e stabila în raport cu datele inițiale.

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 + \varepsilon \end{cases}$$

not $z(x, \varepsilon)$ sol ec
 $y(x)$ sol ec inițială

para

$$\begin{aligned} z(x, \varepsilon) - y(x) &= z(x, \varepsilon) - y(x) + (z(c, \varepsilon) - y(c)) = y_0 + \varepsilon - y_0 + f(c, z) - f(c, y) \\ \text{T Lagrange în raport cu a 2-a variabilă} \\ z(x, \varepsilon) - y(x) &= \varepsilon + \frac{\partial f}{\partial y}(c, c_1)(z(x, \varepsilon) - y(x)) \end{aligned}$$

$$z'(x, \varepsilon) = f(x, z(x, \varepsilon))$$

$$y'(x) = f(x, y)$$

$$z'(x, \varepsilon) - y'(x) = \frac{\partial f}{\partial y}(z(x, \varepsilon) - y(x)) + \frac{(z(x, \varepsilon) - y(x))^2}{2} \frac{\partial^2 f}{\partial y^2}(c, \eta)$$

$$\Rightarrow z' - y' = \frac{\partial f}{\partial y}(z - y)$$

$$\text{not } u = z - y$$

$$u' = \frac{\partial f}{\partial y}(u)$$

$$u' - u \frac{\partial f}{\partial y} = 0$$

$$u(x_0) = z(x_0, \varepsilon) - y(x_0) = y_0 + \varepsilon - y_0 = \varepsilon$$

$$\frac{u'}{u} = \frac{\partial f}{\partial y}$$

$$\ln u = \int_{x_0}^x \frac{\partial f}{\partial y} dt + C \Rightarrow u = e^C \cdot e^{\int_{x_0}^x \frac{\partial f}{\partial y} dt}$$

$$u(x_0) = \varepsilon \Rightarrow e^C = \varepsilon$$

$$u(x) = \varepsilon e^{\int_{x_0}^x \frac{\partial f}{\partial y} dt}$$

$$z(x, \varepsilon) - y(x) = \varepsilon e^{\int_{x_0}^x \frac{\partial f}{\partial y} dt}$$

$\frac{\partial f}{\partial y} > 0 \Rightarrow$ problema e prost condiționată

$\frac{\partial f}{\partial y} < 0 \Rightarrow$ problema e bine condiționată

I Metode analitice (soluția ecuației sub formă de serie)

II Metode numerice

$$y' = f(x, y) \Rightarrow y \in C^\infty [J]$$

$$y_0 = y(x_0)$$

$$y(x) = \sum_{n=0}^{\infty} \frac{y^{(n)}(x_0)}{n!} (x-x_0)^n$$

$$y(x_0) = y_0$$

$$y'(x_0) = f(x_0, y_0)$$

$$y'' = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f(x, y)$$

$$y''' = \frac{\partial^2 f}{\partial x^2} + 2 \frac{\partial^2 f}{\partial x \partial y} f(x, y) + \frac{\partial^2 f}{\partial y^2} f(x, y) f(x, y) + \left(\frac{\partial f}{\partial y} \cdot \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} f(x, y) \frac{\partial f}{\partial y} \right)$$

$$y' + xy = 0$$

$$y(x_0) = 0$$

$$y'(0) = 1$$

$$y''(0) = 0$$

$$y^{(k+2)}(x) + xy^{(k)}(x) + ky^{(k-1)}(x) = 0$$

$$x=0 \quad y^{(k+2)}(0) = -ky^{(k-1)}(0)$$

$$y^{(3m)}(0) = 0$$

$$y^{(3m+1)}(0) = (-1)^m \prod_{p=1}^m (3p-1)$$

$$y^{(3m+2)}(0) = 0$$

$$k=3p-2: \quad y^{(3p)}(0) = -(3p-2)y^{(3p-2)}(0)$$

inductive

$$y^{(3m)}(0) = 0$$

$$y^{(3m+3)}(0) = -(3m+1)y^{(3m)}(0)$$

$$k=3p-1: \quad y^{(3p+1)}(0) = -(3p+1)y^{(3p-1)}(0)$$

$$\prod_{p=1}^m y^{(3p)}(0) = (-1)^m \prod_{p=1}^m (3p-1) \prod_{p=1}^m y^{(3p-2)}(0)$$

$$y^{(3m+1)}(0) = (-1)^m \prod_{p=1}^m (3p-1)$$

$$k=3p \quad y^{(3p+2)}(0) = -3p y^{(3p-1)}(0) \Rightarrow y^{(3p+2)}(0) = 0$$

$$\Rightarrow y(x) = \sum_{n=0}^{\infty} (-1)^n \frac{n}{\prod_{p=1}^{3p-1} (3p-1)} \frac{x^{3n+1}}{(3n+1)!} \quad (\text{soluție formală})$$

raza de convergență:

$$R = \lim_{n \rightarrow \infty} \frac{|a_n|}{|a_{n+1}|} = \lim_{n \rightarrow \infty} \frac{\frac{n}{\prod_{p=1}^{3p-1} (3p-1)}}{\frac{n+1}{\prod_{p=1}^{3n+3} (3p-1)}} = \lim_{n \rightarrow \infty} \frac{(3n+3)(3n+2)}{(3n+1)} = \infty$$

Metode numerice unipass

$$y' = f(x, y), \quad a = x_0, \quad b = x_{n+1}, \quad [a, b]$$

$$x_k = x_0 + kh \quad \rightarrow \text{pasul (constant)}$$

$$\bar{y}_k \approx y(x_k)$$

$$\bar{y}_{k+1} = \bar{y}_k + g(x_k, \bar{y}_k), \quad \text{metoda unipass}$$

1. Metoda Euler (liniilor poligonale)

$$y_k = y(x_k) \text{ soluția exactă}$$

$$\bar{y}_k - \text{soluția aproximativă}$$

$$y' = f(x, y) \Rightarrow y(k+1) - y(x_k) = \int_{x_k}^{x_{k+1}} f(x, y(x)) dx \approx (x_{k+1} - x_k) f(x_k, y(x_k))$$

$$\bar{y}_{k+1} = \bar{y}_k + h f(x_k, \bar{y}_k)$$

$$\bar{y}_1 = y_0 + h f(x_0, y_0)$$

$$|y_k - \bar{y}_k|$$

$$|\bar{y}_{k+1} - y_{k+1}| = |\bar{y}_k + h f(x_k, \bar{y}_k) - y_{k+1}| = |\bar{y}_k + h f(x_k, \bar{y}_k) - y_k - \int_{x_k}^{x_{k+1}} f(t, y(t)) dt|$$

$$\leq |\bar{y}_k - y_k| + |h f(x_k, \bar{y}_k) - \int_{x_k}^{x_{k+1}} f(t, y(t)) dt| \stackrel{\text{f de medie pt integrala}}{=} |\bar{y}_k - y_k| + |h f(x_k, \bar{y}_k) - h f(\xi_k, y(\xi_k))|$$

$$\leq |\bar{y}_k - y_k| + hK(\bar{y}_k - y_k)$$

$$|y_{k+1} - \bar{y}_{k+1}| \leq |y_k - \bar{y}_k| (1 + h_k k)$$

$$\text{not } l_k = f(x_k - \bar{y}_k)$$

$$l_{k+1} \leq l_k (1 + h_k k)$$

$$\prod_{k=0}^{n-1} l_{k+1} \leq \left(\prod_{k=0}^{n-1} l_k \right) (1 + h_k k)^n$$

$$l_n \leq e_0 (1 + h_k k)^n$$

$$|y_m - \bar{y}_m| \leq |y_0 - \bar{y}_0| (1 + h_k k)^n$$

rel. care evaluează eroarea în cazul metodei lui Euler

$$y(x_{k+1}) - y(x_k) \approx h M \rightarrow \text{ordin de aproximare 1}$$

2. Metode de tip Runge-kutta

$$y' = f(x, y), \quad \overset{\text{metodă}}{k_0 = h f(x_0, y_0)}$$

$$y(x_0) = y_0 \quad k_1 = h f(x_0 + h, y_0 + b_{1,0} k_0)$$

$$k_p = h f(x_0 + a_p h, y_0 + \sum_{j=0}^{p-1} b_{pj} k_j)$$

$$(*) \bar{y}_{m+1} = \bar{y}_m + \sum_{k=0}^p A_k \cdot k_k$$

Def. Metoda (*) se numește metoda Runge-kutta de ordin p dacă dezvoltarea în serie Taylor a lui $\bar{y}_{m+1}$ în jurul punctului x_m are primii p termeni aceiași cu cei din soluția exactă considerând ca $\bar{y}_m = y_m$.

$$p=0 \quad \bar{y}_{m+1} = \bar{y}_m + A_0 k_0$$

$$\bar{y}_{m+1} = \bar{y}_m + A_0 h f(x_m, \bar{y}_m)$$

$$y(x_{m+1}) = y(x_m + h) = y(x_m) + h y'(x_m) + \dots = y(x_m) + h f(x_m, y_m) + \dots \Rightarrow A_0 = 1,$$

$$\bar{y}_{m+1} = \bar{y}_m + h k_0$$

$$p=1: \bar{y}_{n+1} = \bar{y}_n + A_0 k_0 + A_1 k_1$$

$$\bar{y}_{n+1} = \bar{y}_n + A_0 h f(x_n, \bar{y}_n) + A_1 h f(x_n + A_1 h, \bar{y}_n + b_{1,0} h f(x_n, \bar{y}_n)) = g(h)$$

$$g(0) = \bar{y}_n$$

$$g'(h) = A_0 f(x_n, \bar{y}_n) + A_1 [f(x_n + A_1 h, \bar{y}_n + b_{1,0} h f(x_n, \bar{y}_n)) + h \frac{\partial f}{\partial x} a_1 + \frac{\partial f}{\partial y} (\bar{y}_n + b_{1,0} h f(x_n, \bar{y}_n))]]$$

$$g'(0) = A_0 f(x_n, \bar{y}_n) + A_1 [f(x_n, \bar{y}_n) + \frac{\partial f}{\partial y} (\bar{y}_n + b_{1,0} h f(x_n, \bar{y}_n))]]$$

$$y(x_{n+h}) = y(x_n + h) = y(x_n) + h f(x_n, y_n) + \frac{h^2}{2} \left[\frac{\partial f}{\partial x} (x_n, y_n) + f(x_n, y_n) \frac{\partial f}{\partial y} (x_n, y_n) \right]$$

$$\begin{cases} A_0 f(x_n, y_n) + A_1 f(x_n, y_n) = f(x_n, y_n) & | : f(x_n, y_n) \\ A_1 \cdot \frac{\partial f}{\partial x} a_1 + \frac{\partial f}{\partial y} b_{1,0} f(x_n, y_n) = \frac{1}{2} \left[\frac{\partial f}{\partial x} (x_n, y_n) + f(x_n, y_n) \frac{\partial f}{\partial y} (x_n, y_n) \right] \end{cases}$$

$$\begin{cases} A_0 + A_1 = 1 & (A_0 = A_1 = \frac{1}{2}) \\ A_1 a_1 = \frac{1}{2} & a_1 = 1 \\ A_1 b_{1,0} = \frac{1}{2} & b_{1,0} = 1 \end{cases}$$

$$\bar{y}_{n+1} = \bar{y}_n + \frac{1}{2} (k_0 + k_1) \quad \text{metoda cea mai flosie}$$

$$\begin{matrix} p=2 \\ p=3 \end{matrix} \quad \text{tema (sub examen)}$$

Metode multiple

$$\bar{y}_{n+1} = \bar{y}_n + \int_{x_n}^{x_{n+1}} f(t, y(t)) dt$$

$$x_{n-m} < x_{n-m+1} < \dots < x_n$$

$$\bar{y}_{n+1} = \bar{y}_n + \sum_{k=n-m}^n A_k f(x_k, \bar{y}_k) \quad \text{f. cuadratura de tip deschis.}$$

$$\bar{y}_{n+1} = \bar{y}_n + \sum_{k=n}^{n+p} A_k f(x_k, \bar{y}_k) \quad \text{f. cuadratura de tip inchis.}$$

$$\bar{y}_{n+1} = \bar{y}_n + \sum_{k=n-m}^{n+1} A_k f(x_k, \bar{y}_k)$$

↳ (f. de tip predictor-corrector)

$$\bar{y}_{n+1} = \bar{y}_n + h f(x_n, \bar{y}_n)$$